







Mapping/Modelling Biodiversity

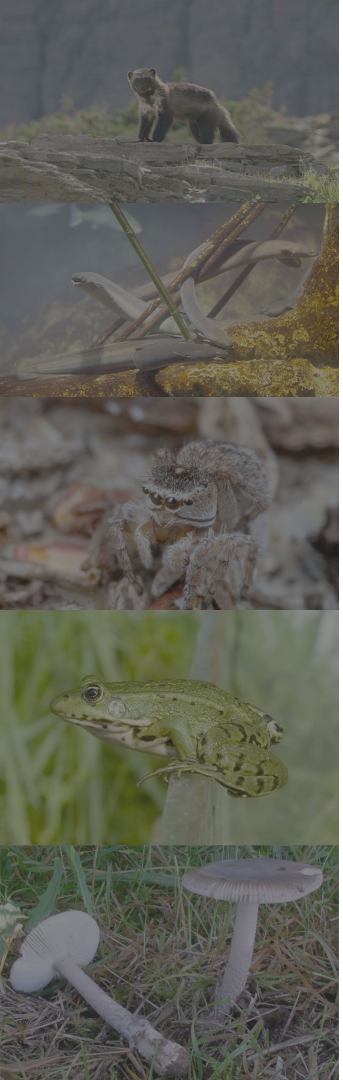
Using GBIF data to conserve nature

Sam Wenaas Perrin

GJÆREVOLL CENTRE
Biodiversity Foresight Analyses

NINA
Norwegian Institute for Nature Research

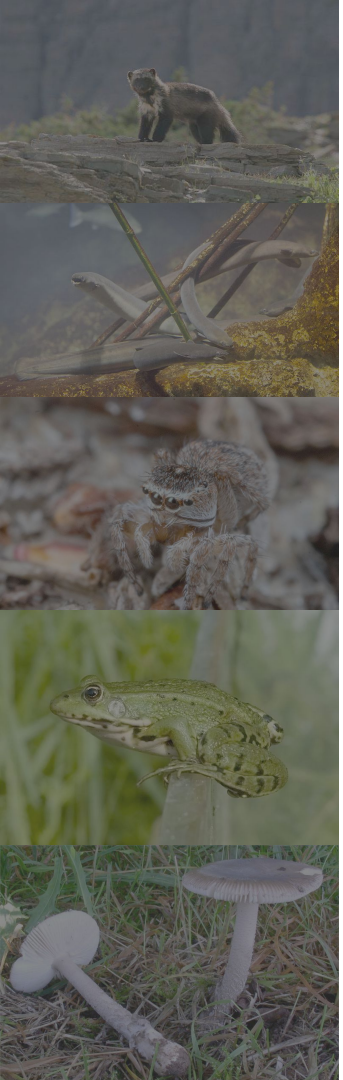
UNIVERSITETET I BERGEN



Why is mapping biodiversity so important?

How do we map biodiversity?

How does GBIF help?



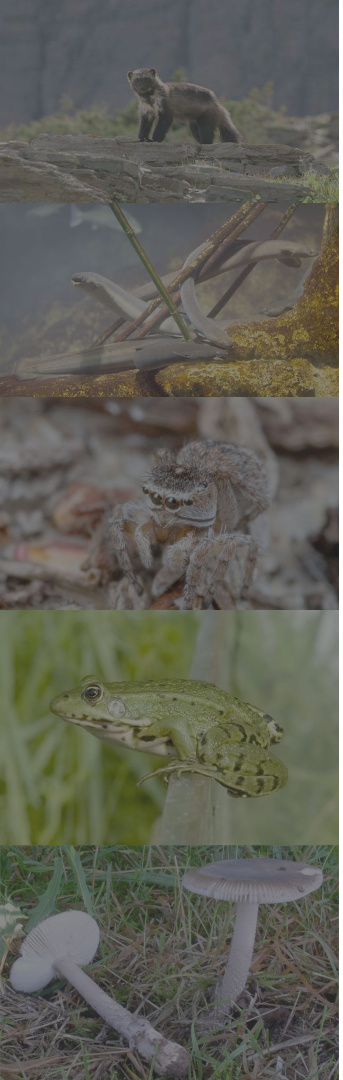




THE UNIVERSITY OF MELBOURNE

University of Bremen

Image credit: Reef - Workfortravel (CC BY-SA 4.0)

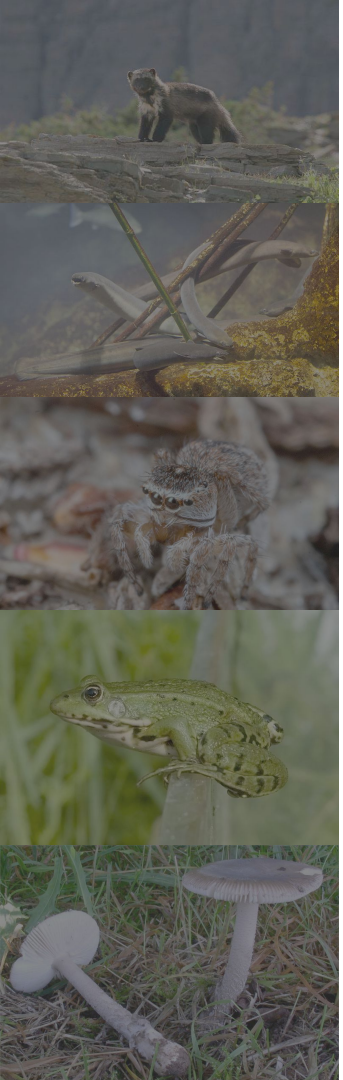


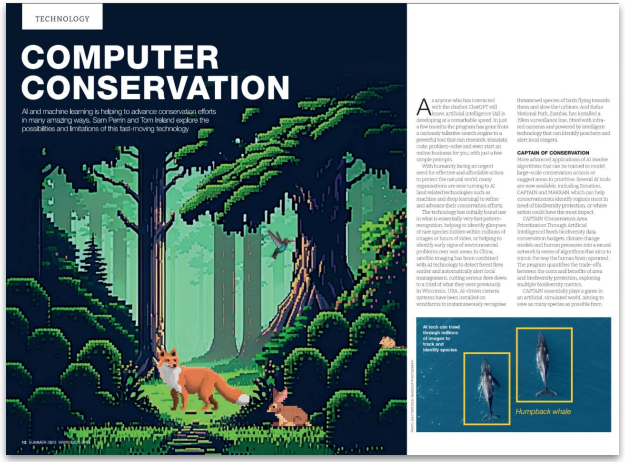
NTNU
Vitenskapsmuseet

cbd
CENTRE FOR BIODIVERSITY DYNAMICS

Ducky

GJÆREVOLLSENTERET
Framtidsanalyser av naturmangfold



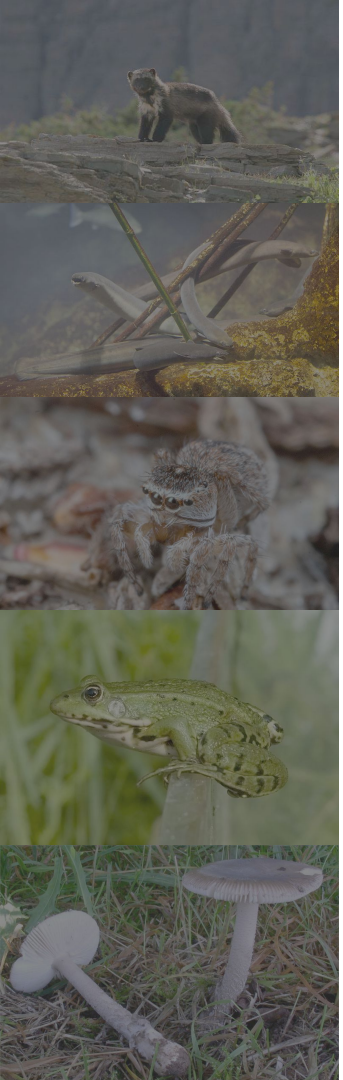

TECHNOLOGY
COMPUTER CONSERVATION
AI and machine learning is helping to advance conservation efforts in many amazing ways. Sam Perrin and Tom Ireland explore the possibilities and limitations of this fast-moving technology
CAPTAIN OF CONSERVATION
AI tech can trawl through millions of images to track and identify species
Humpback whale


FEATURE
ALIENS & INVADERS & EXOTICS, OH MY
THE LANGUAGE OF INVASIVE BIOLOGY


BIOLOGISTS OF MIDDLE EARTH


KAIJU RAMEN
MAGAZINE
KRM FOOD JAN 2023

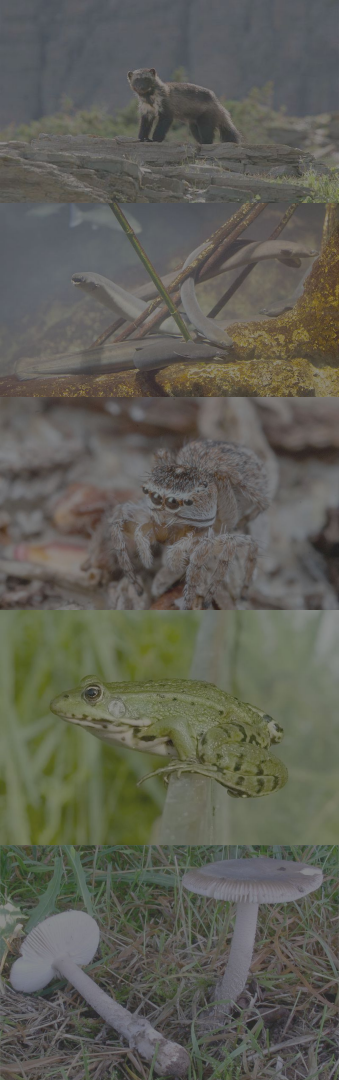

The Biologist
Royal Society of Biology
FEATURES
INTERVIEWS
COVID-19
MEET OUR MEMBERS
OPINION
BOOK REVIEWS
ARCHIVE
FEATURES
The way of the jackal
Biography
Dr Sam Wenaas Perrin is an invasion ecologist and climate data analyst and founder/editor of the blog Ecology for the Masses.
When the golden jackal appeared most onlookers assumed it was just a matter of time before the species was duly assigned a classification by local authorities. Yet first Finland then Norway announced that the species wouldn't make it on to the list



Input

GBIF

Output



Hotspots
GJÆREVOLLSENTERET · NTNU

Om prosjektet | Utforsk | Kodebase | Folk | NO | EN

Utforsk biodiversitetskart frå Hotspot-prosjektet. Vel eit kartlag i lista for å vise det på kartet. Karta er tilgjengelege som Cloud Optimized GeoTIFF og kan lastast ned enkeltvis eller brukast direkte i GIS-programvare.

Kartlag

TAKSONOMISK GRUPPE

Karplanter

KARTTYPE

Artsrikdom

ARTSGRUPPE

Alle artar

AKTIVT LAG

Artsrikdom (karplanter)

Låg — Høg

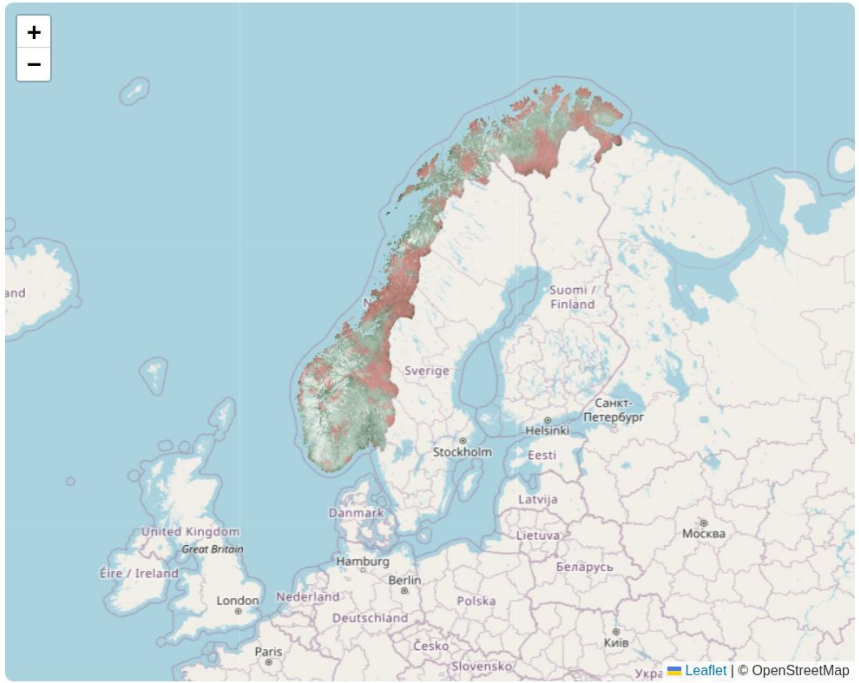
Vis usikkerheit

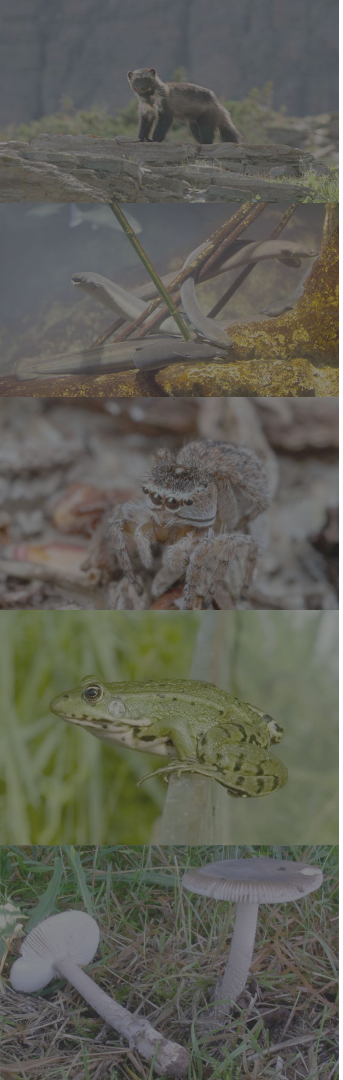
USIKKERHEIT

Middels usikkerheit

Høg usikkerheit

Last ned
allSpeciesUploadRaster.tif
14.8 MB

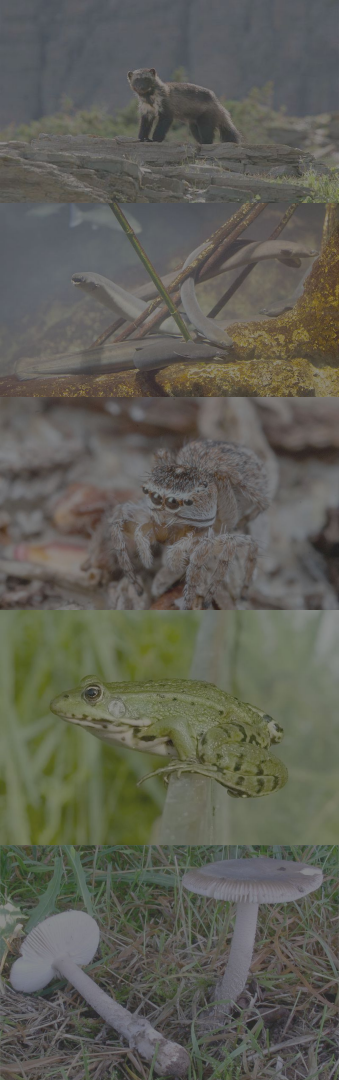




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How do we map biodiversity?

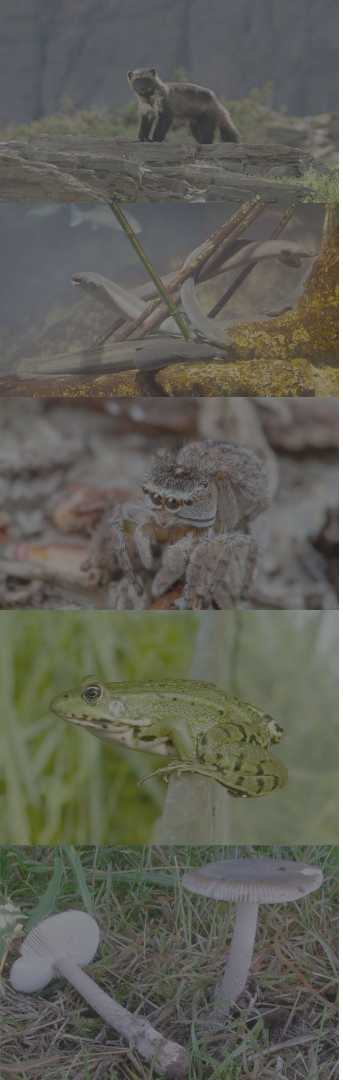
How does GBIF help?

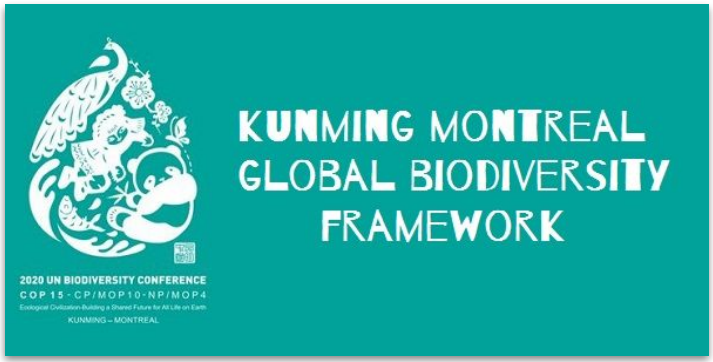


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…the twenty-three targets to be achieved by 2030 include **30 per cent conservation of land, sea and inland waters**, 30 per cent restoration of degraded ecosystems…

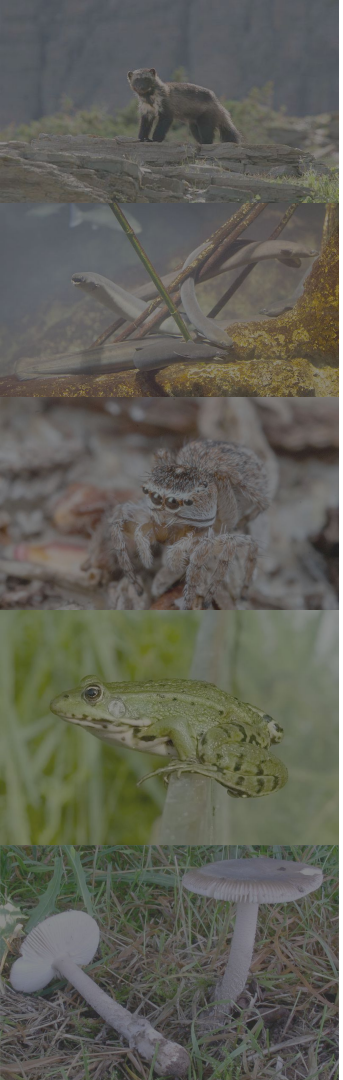
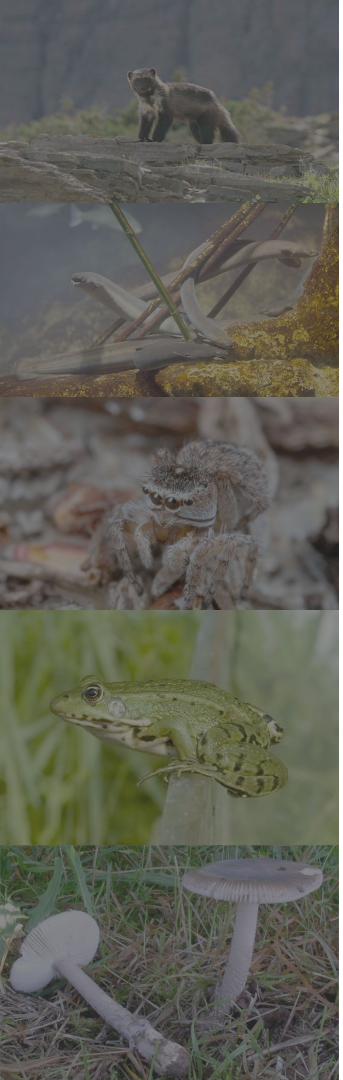






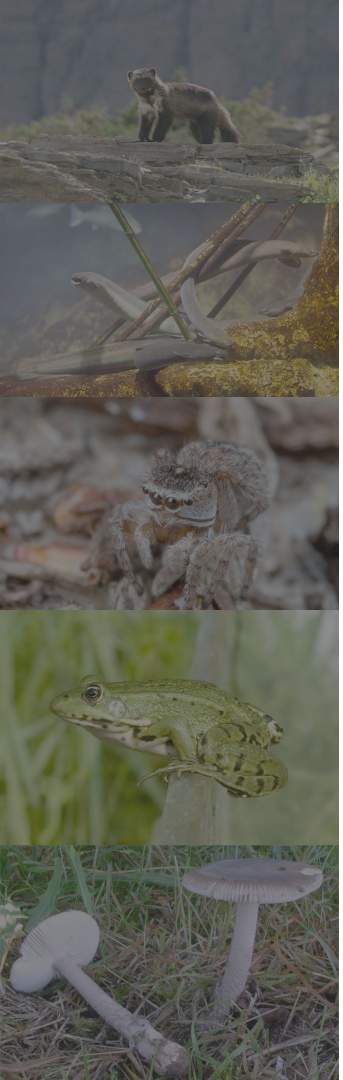
Image credit (wind farm): Statkraft (CC BY-NC-ND 2.0)



You need to understand which areas are the most valuable

What is value

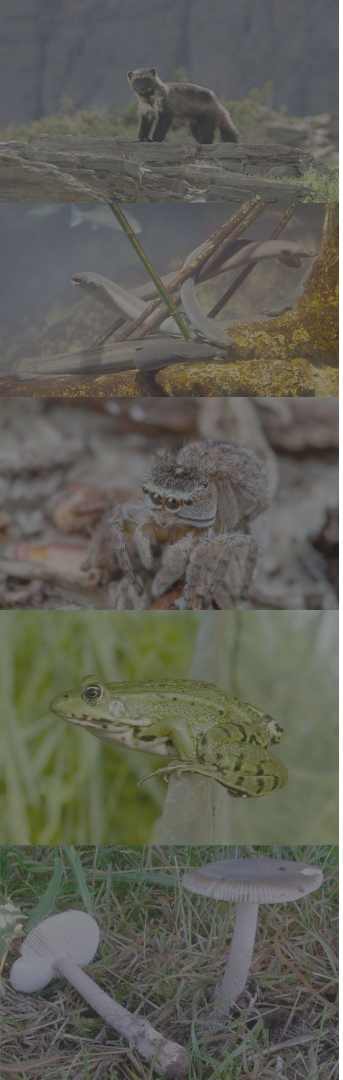
You need to understand the spatial distribution of species



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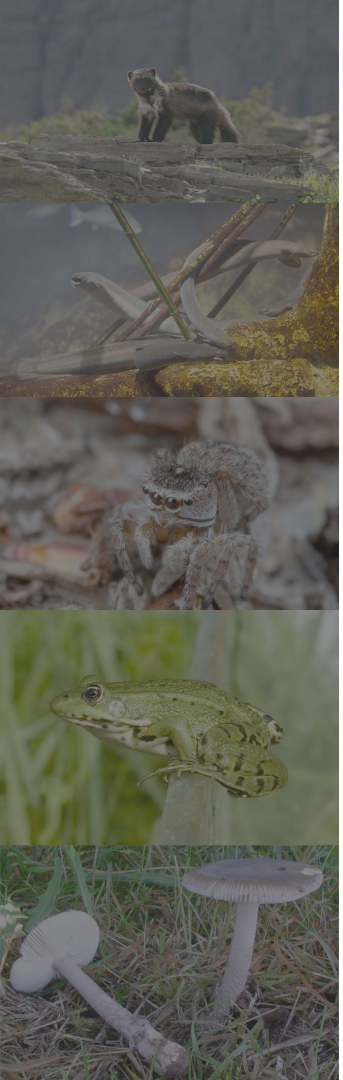
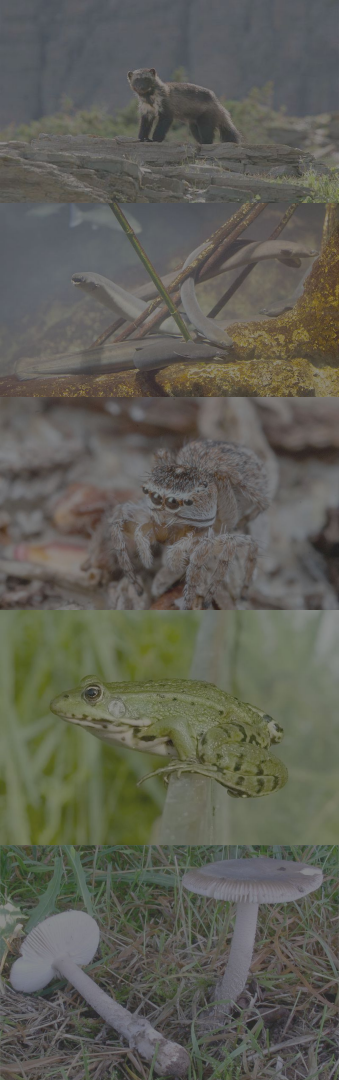
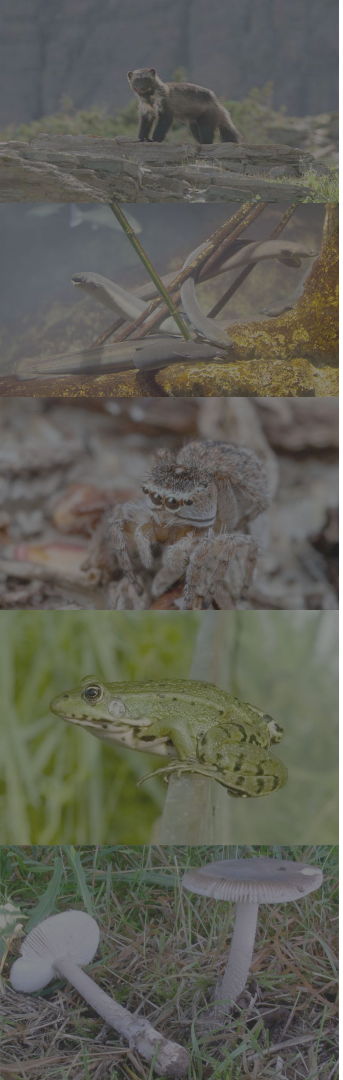




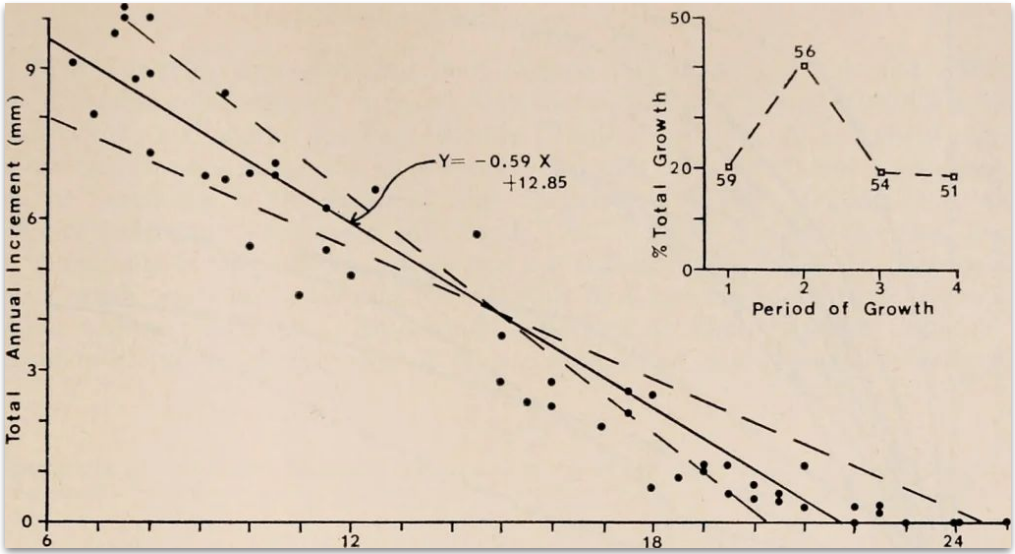
Image credit: NTNU Vitenskapsmuseet

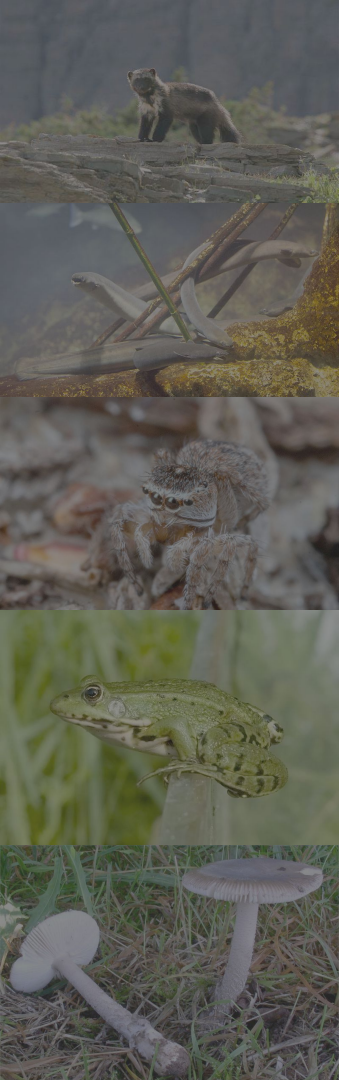


We need to use **the areas we have data for** to extrapolate and make predictions about the areas **we don’t have data for**



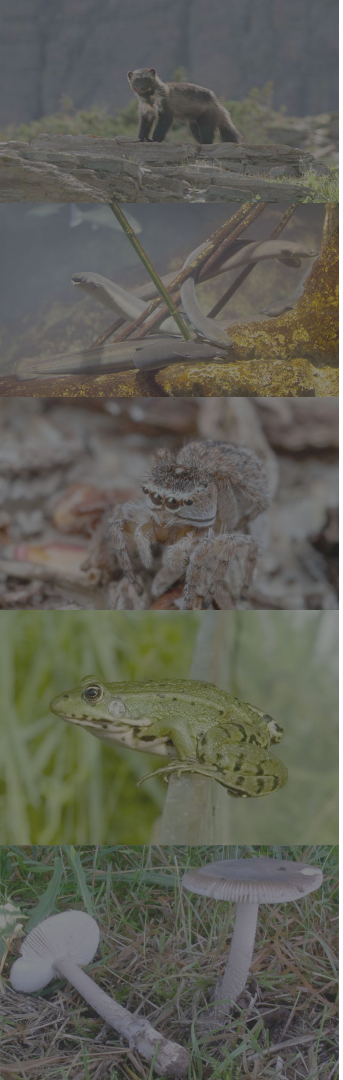
Ecological Modelling





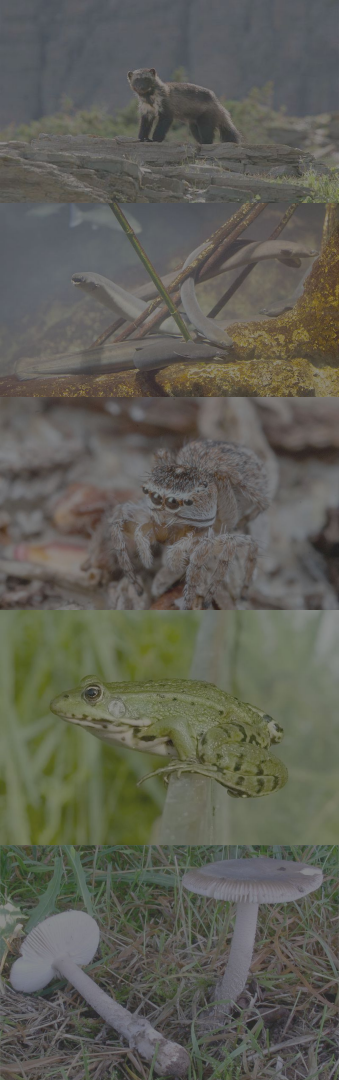
What do you need for a **Species Distribution Model**?

- Species occurrence data
- Environmental data
- Additionally:
  - Spatial data
  - Trait/phylogenic data



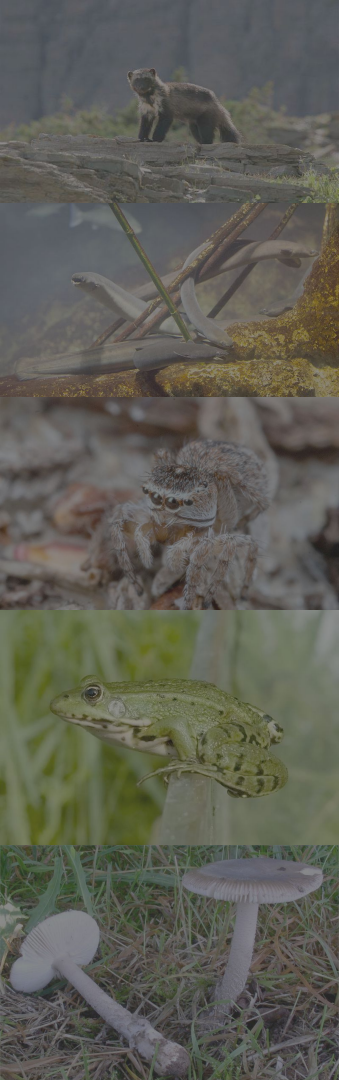
Decisions you need to make

- What are you measuring?
- What's your output? Will you use it for predictions? Inference about a species?
- Who will be using your results?
- How will you show uncertainty?
- What's the scale of your model/map?



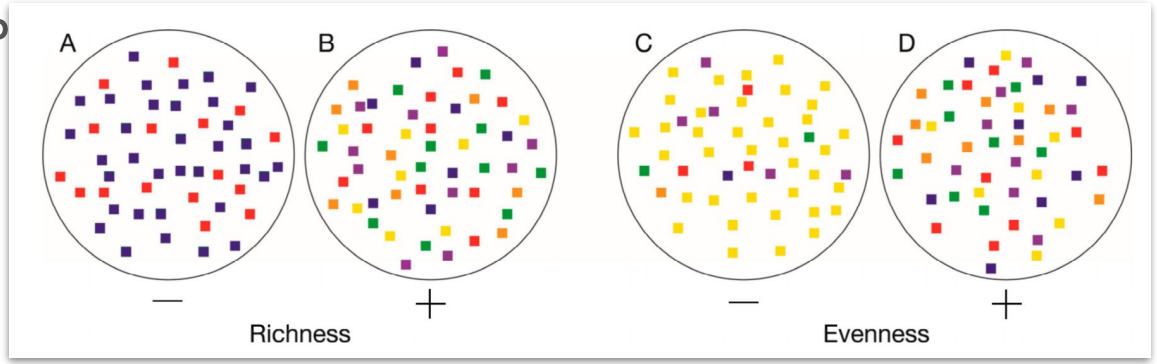
How can we measure biodiversity?

- **Species richness/evenness**
- Rare/threatened species richness
- Genetic diversity
- Functional richness
- Phylogenetic richness

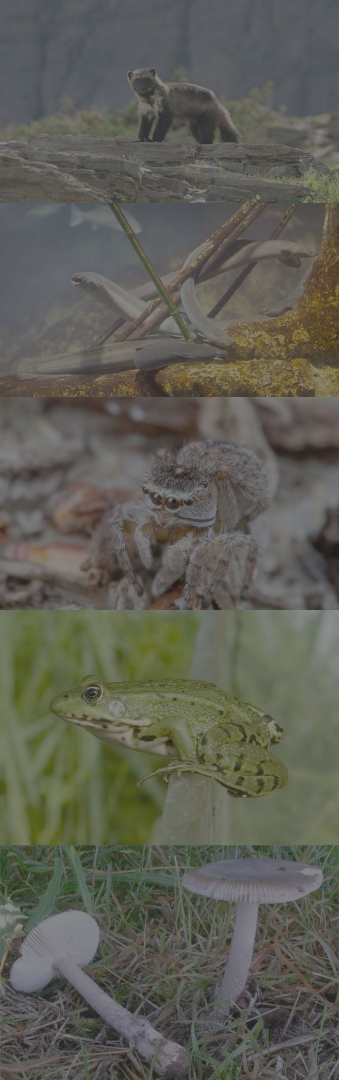


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- P



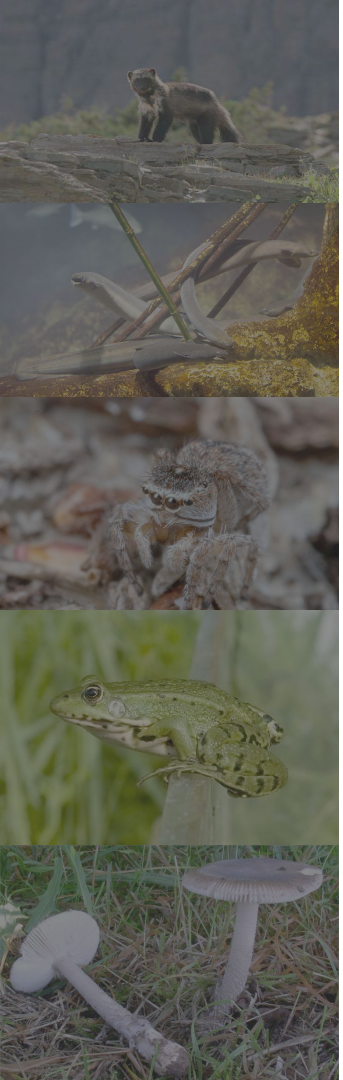
LaFevor & Pitts, 2022



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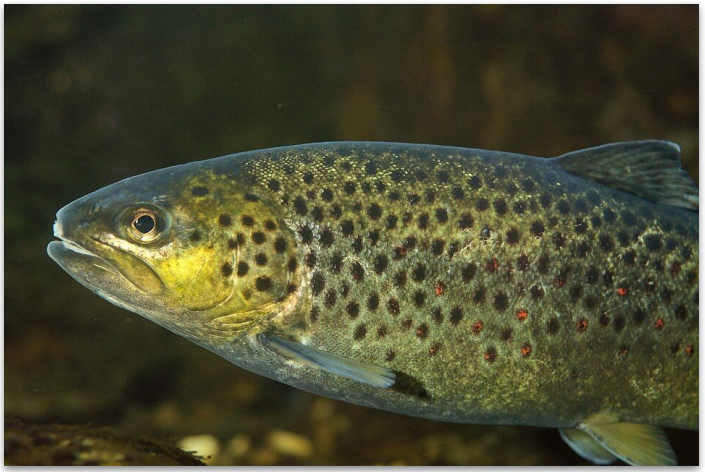
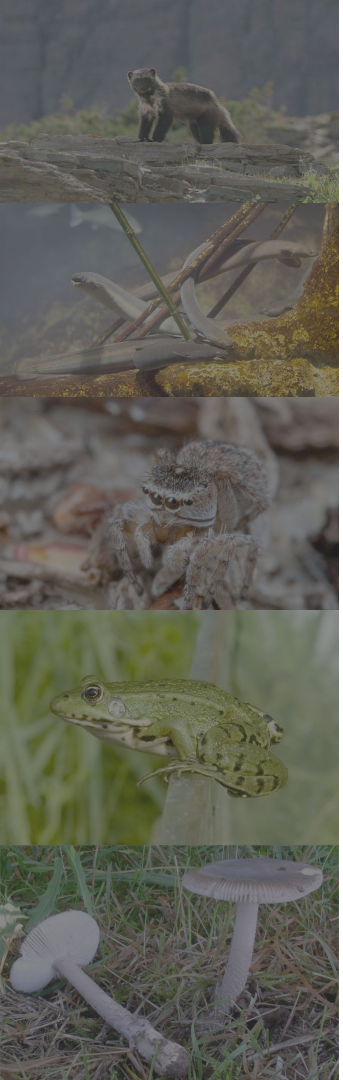


Image credit: Gilles San Martin (CC BY-SA 2.0)



UNCERTAINTY



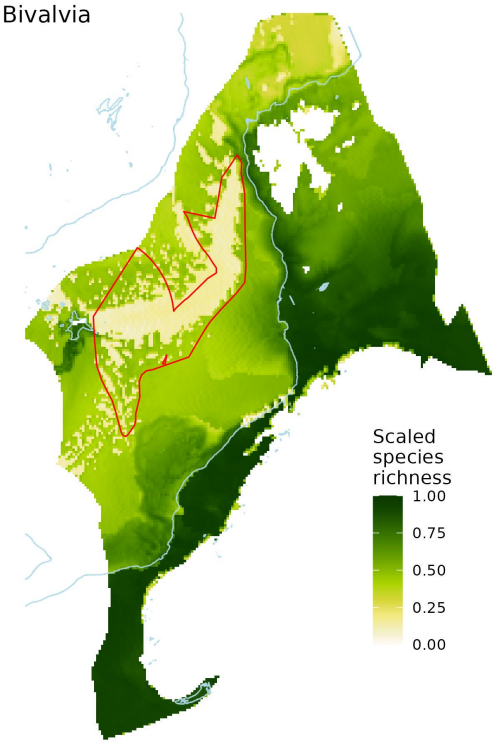




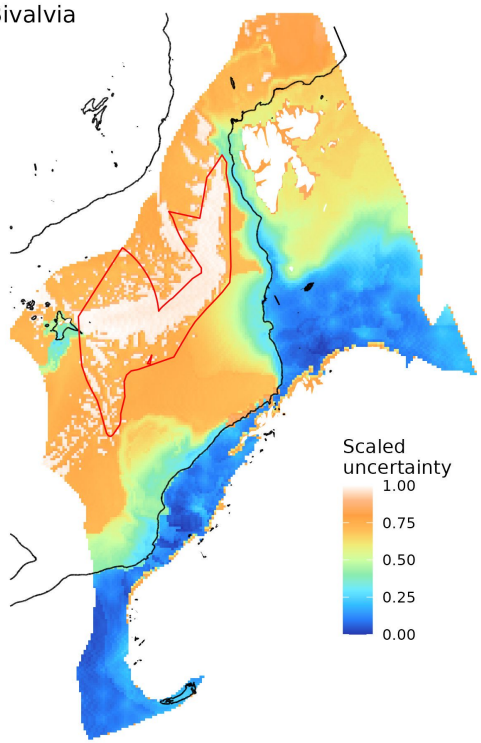


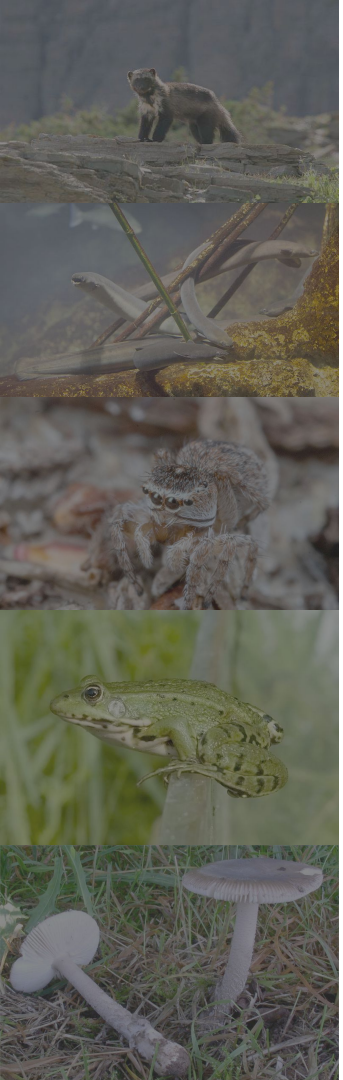


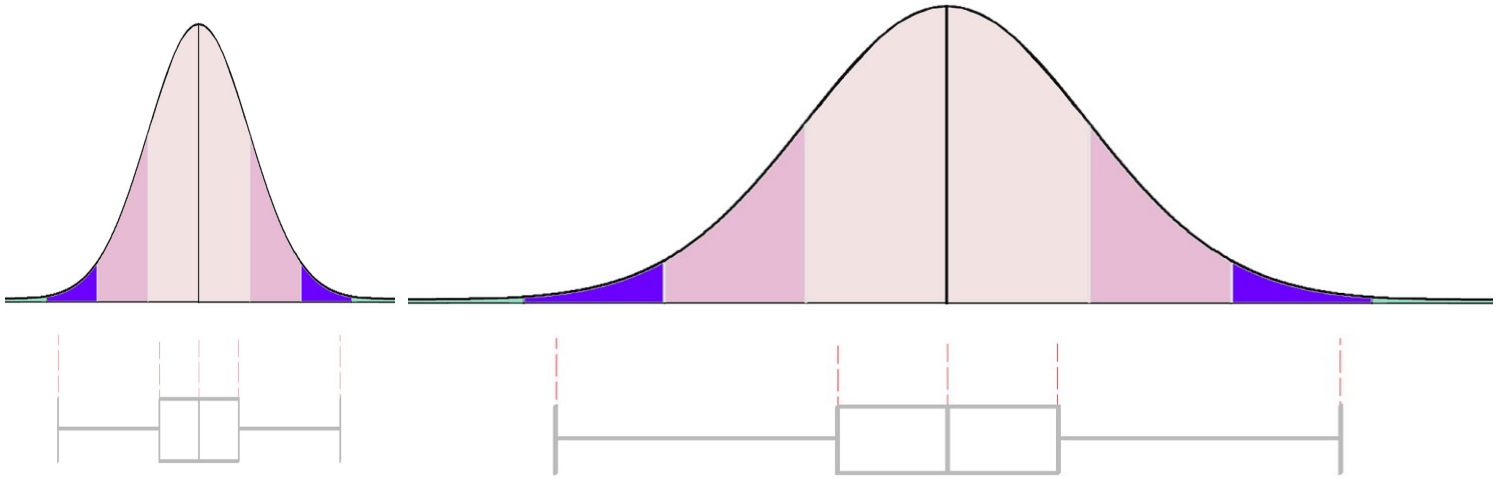
Bivalvia



Bivalvia



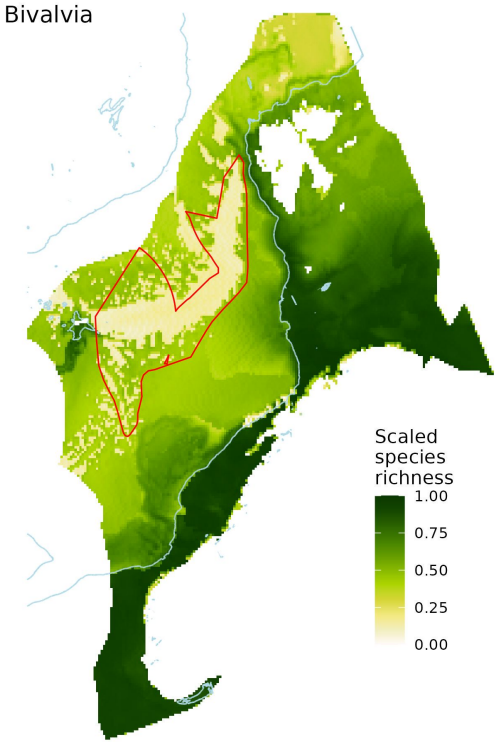


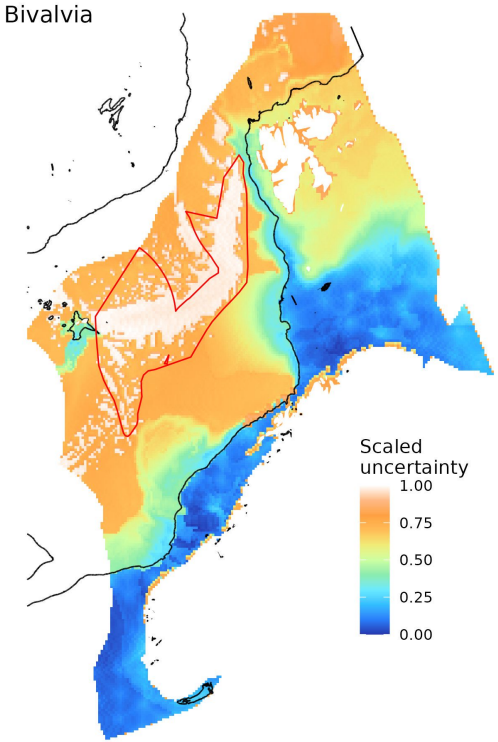








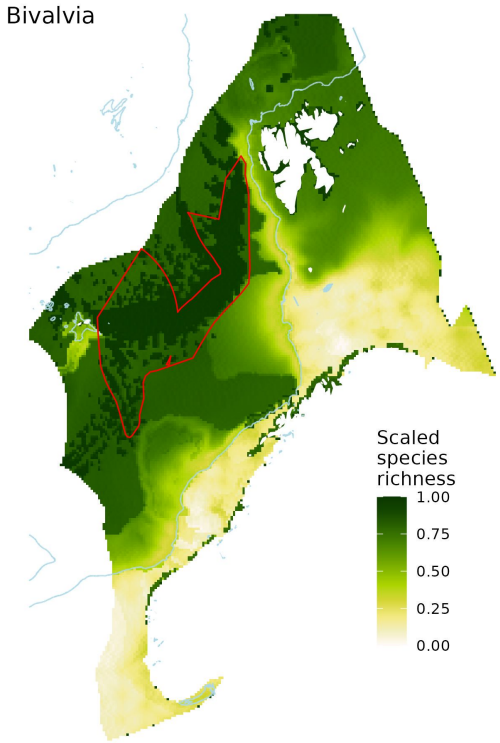

Bivalvia
Scaled species richness
1.00
0.75
0.50
0.25
0.00

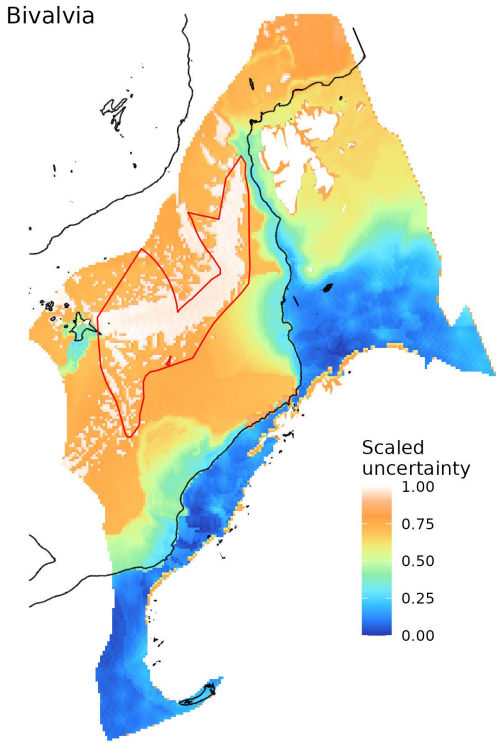

Bivalvia
Scaled uncertainty
1.00
0.75
0.50
0.25
0.00








Bivalvia
Scaled species richness
1.00
0.75
0.50
0.25
0.00


Bivalvia
Scaled uncertainty
1.00
0.75
0.50
0.25
0.00

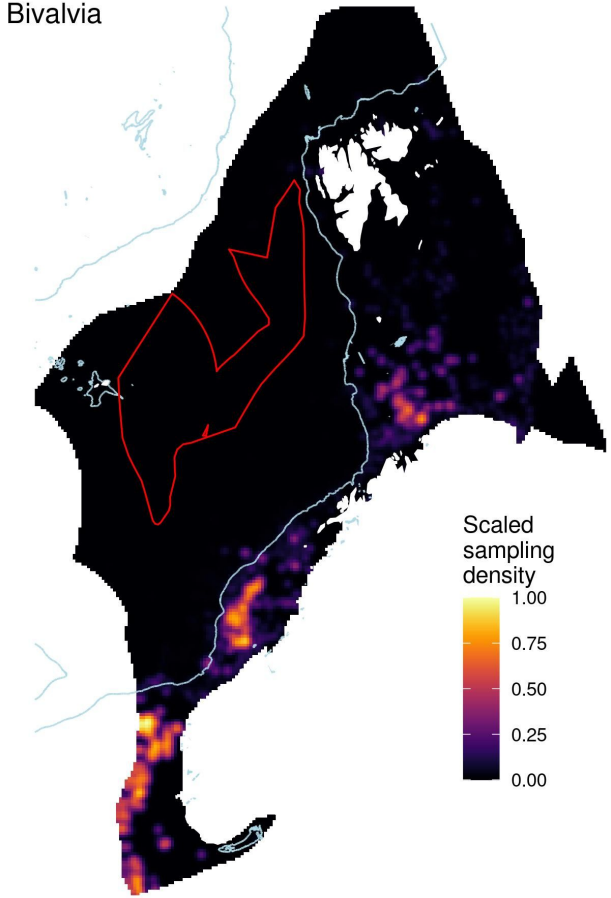


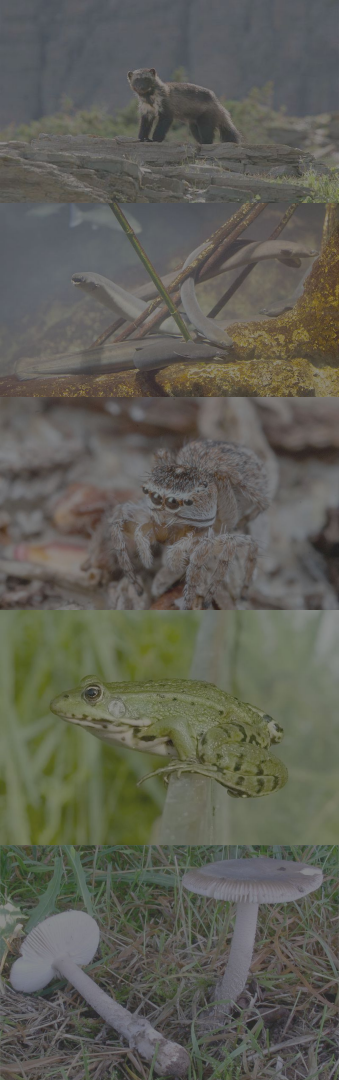




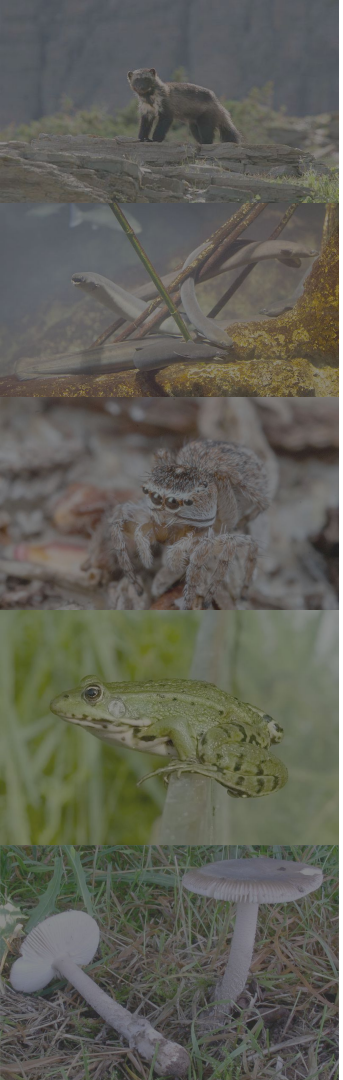




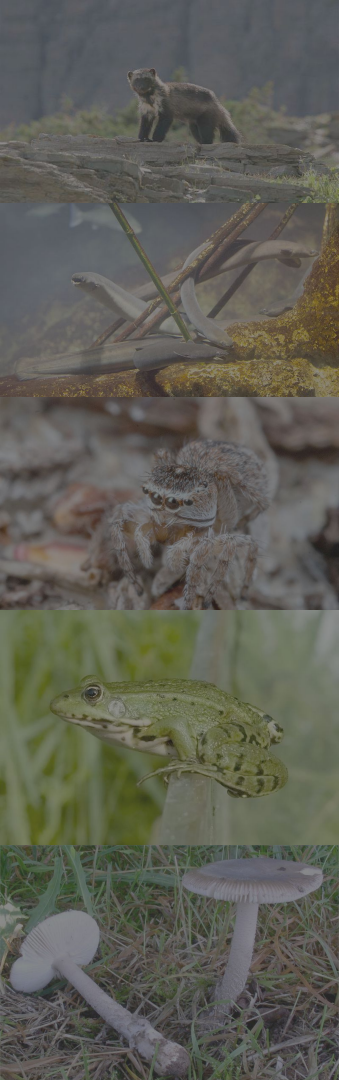

Bivalvia
Scaled sampling density
1.00
0.75
0.50
0.25
0.00



	Low richness	High richness
Low uncertainty	Useable land	Conservation
High uncertainty	**Surveying**	**Surveying**



TECHNICAL BREAK WHERE WE ALL HOPE THE HOTSPOTS WEBSITE HASN’T CRASHED



SPATIAL AUTOCORRELATION












JAMESON
THE OLD JAMESON DISTILLERY





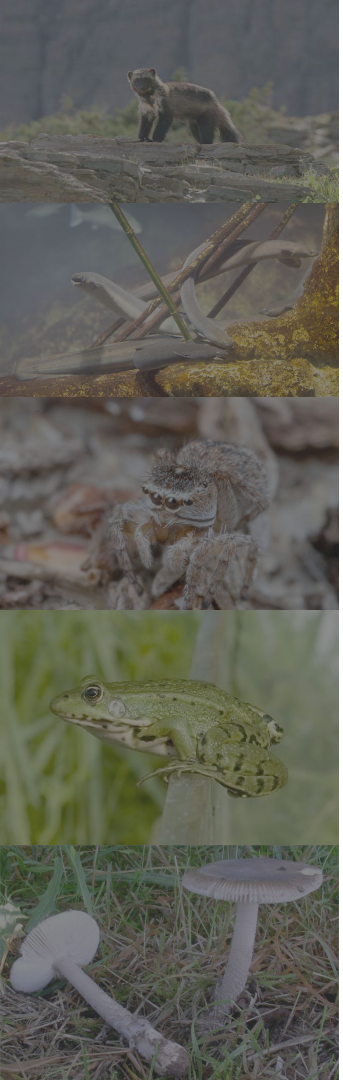







JAMESON
THE OLD JAMESON DISTILLERY

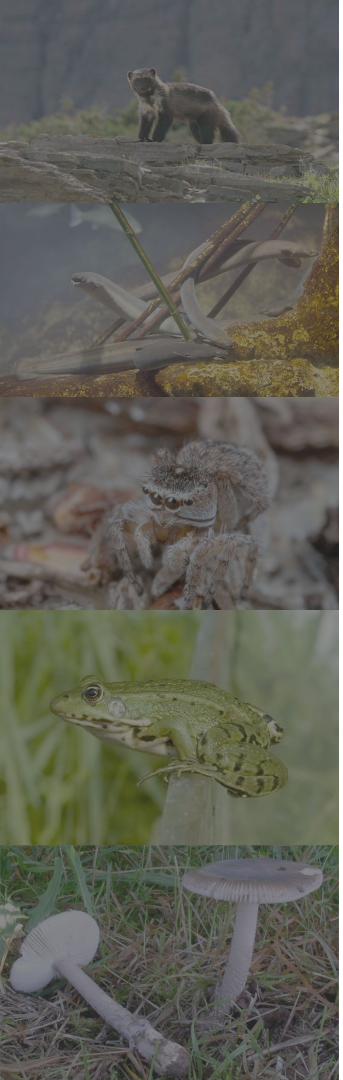




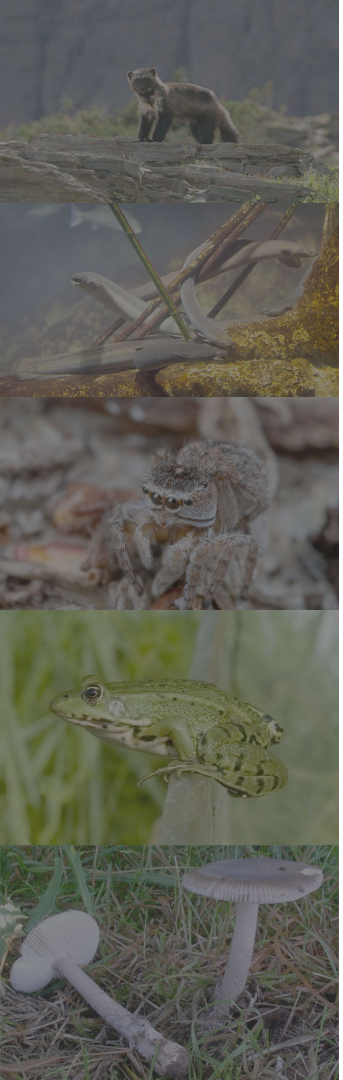
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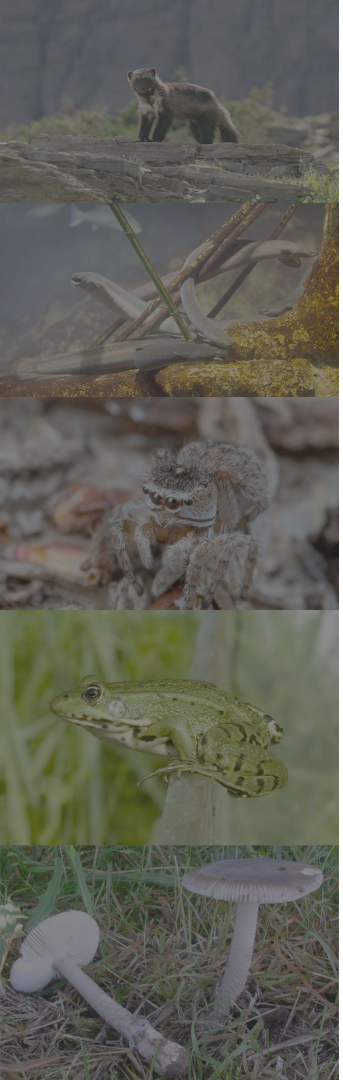
How does GBIF help?







GBIF

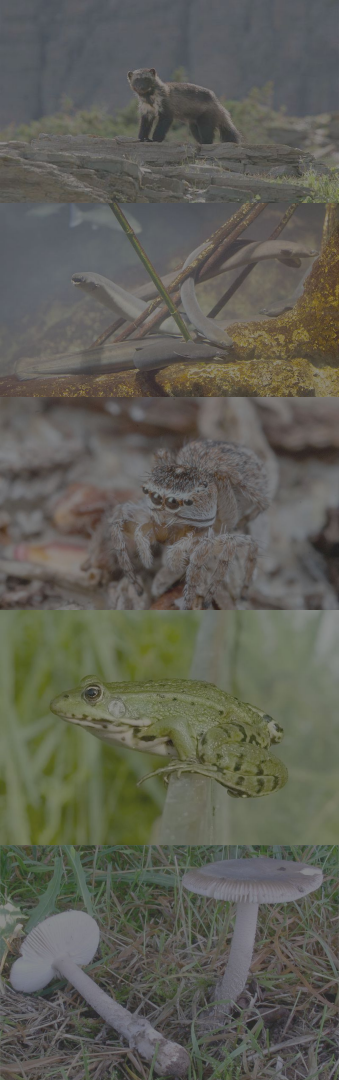


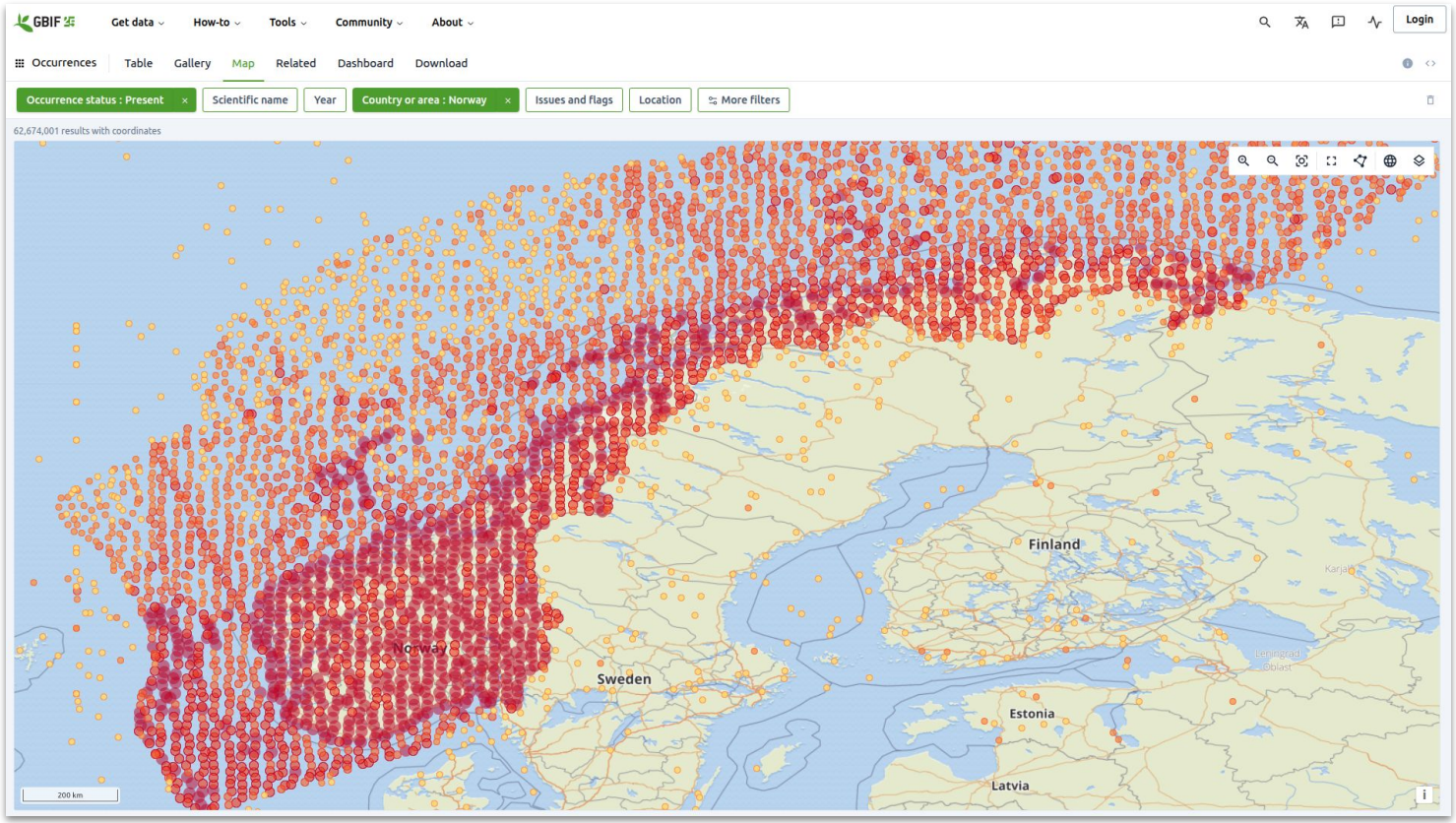
3,846,589,630
Occurrence records

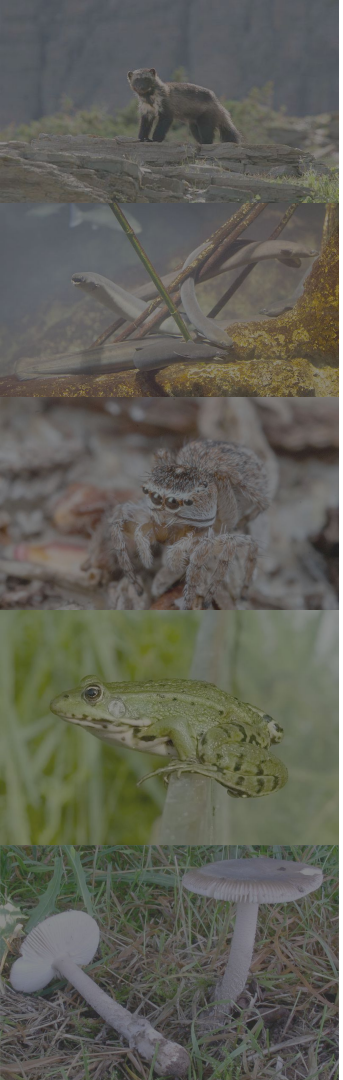
124,809
Datasets

2,817
Publishing institutions

16,018
Peer-reviewed papers using data




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Occurrence status : Present
Scientific name
Year
Country or area : Norway
Issues and flags
Location
More filters
62,674,001 results with coordinates
Finland
Norway
Sweden
Estonia
Latvia
200 km



Dataset
Number of occurrences

538 results

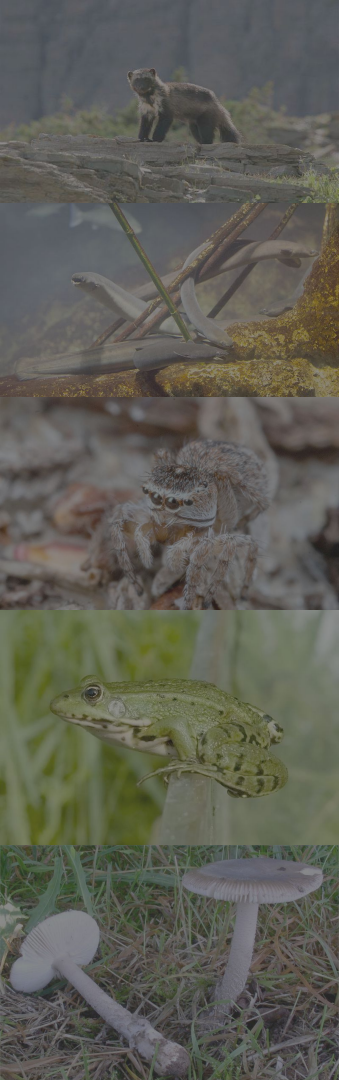
Norwegian Species Observation Service	4,589,614
Vascular Plants, Field notes, Oslo (O) UiO	1,476,412
Vascular Plant Herbarium, Oslo (O) UiO	698,820
Vascular plant field notes, NTNU University Museum	333,543
Biofokus	326,985
Vascular plant herbarium TRH, NTNU University Museum	203,911
Vascular Plant Herbarium, UiB	173,231
Arealrepresentativ naturovervåkning (ANO)	158,552
Vascular plant herbarium, The Arctic University Museum of Norway (TROM)	151,828
Jordal	144,856

Previous Next

Event date



For date ranges, start date is used.
98% of all records have a value



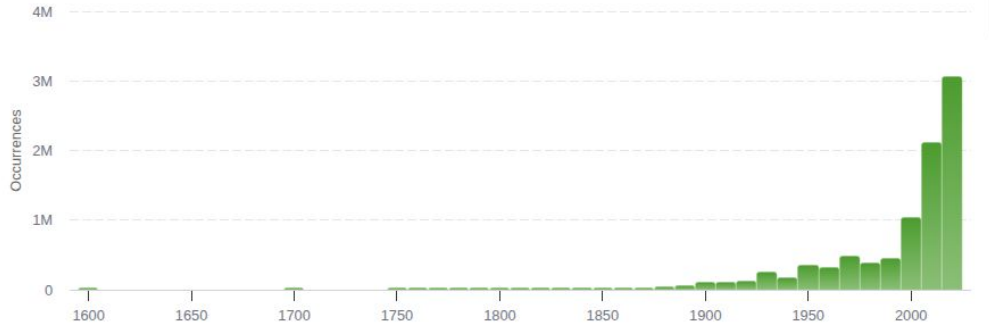
Dataset
Number of occurrences

538 results

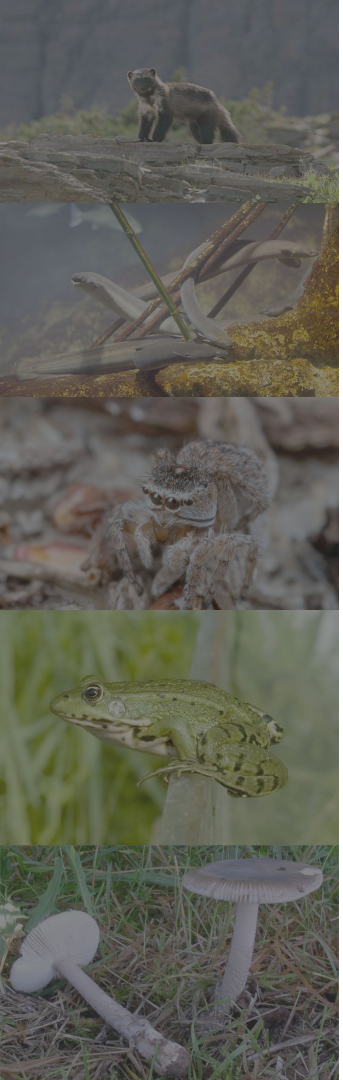
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Previous Next

Event date



For date ranges, start date is used.
98% of all records have a value



Sampling event | Registered 30 July 2012

Vascular plant field notes, NTNU University Museum

Published by Norwegian University of Science and Technology

Licence CC BY 4.0

774 citations | 336,868 occurrences

About | Events | Metrics | Download

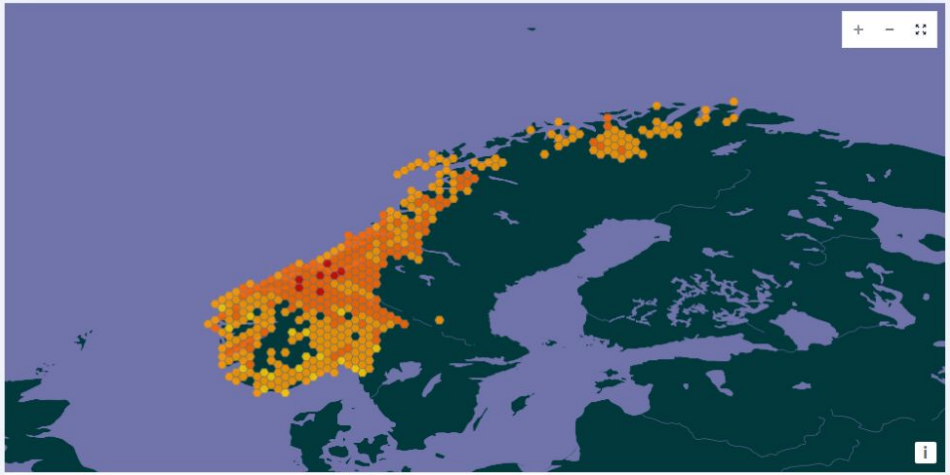
Description

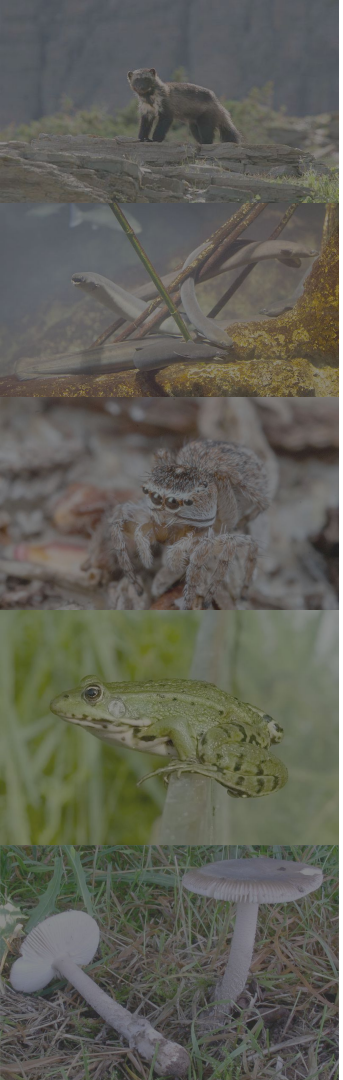
Publication date: 6 November 2025

Network: Living Norway Ecological Data Network

Vascular plant field notes, Norway. Observation of vascular plants from several thousand localities using standardized cross-lists.

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Occurrence metrics

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Methodology

Sampling

Observations of all vascular plant species within a certain area, were recorded in the field using standardized cross-list paper forms covering most vascular plants to be found in Norway. From the late 1990ies onwards, the paper forms have been digitized and the observations recorded in a database. At present the dataset contains observation made until 2006. The area of which each list cover, varies from a few ha to several square km, and the coordinate of each species observation is thus the center point of that area.

Study extent

The dataset is a collection of observations made during various research project, including surveys and shorter field trips, at the NTNU University Museum from the first part of the 1900s and onwards. Included are also a few observation made by skilled amateurs. The dataset contains observation from several thousand localities throughout Norway, and also a few localities in the middle part of Sweden (Dalarna and Värmland).

Quality control

The digitized data have been subjected to thorough proofreading by skilled personnel prior to publication.

Method steps

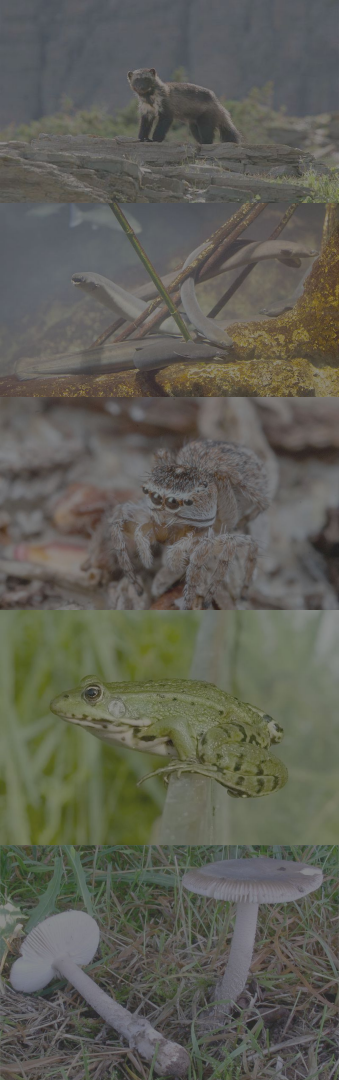
1 No particular procedures

Additional info

A new revised dataset was published in December 2019, including new uuids and revised geographical position and geographical extent of each locality. In 2022 data from an additional 1100 cross-list forms were added.

Contacts

NTNU University Museum	Originator Metadata author
Dag-Inge Øien	Administrative point of contact



Dag-Inge Øien Administrative point of contact

GBIF registration

Registration date	30 July 2012
Metadata last modified	7 November 2025
Publication date	6 November 2025
Hosted by	Norwegian University of Science and Technology
Installation	NTNU
Endpoints	Darwin Core Archive https://gbif.vm.ntnu.no/archive.do?r=vascularplantfieldnotes EML https://gbif.vm.ntnu.no/eml.do?r=vascularplantfieldnotes
Preferred identifier	DOI 10.15468/kkb2x0
Alternative identifiers	https://gbif.vm.ntnu.no/resource?r=vascularplantfieldnotes https://gbif.vm.ntnu.no/ipt/resource?r=vascularplantfieldnotes http://gbif.vm.ntnu.no/ipt/resource?r=vascularplantfieldnotes https://data.gbif.no/ipt/resource?r=trh_vxl http://data.gbif.no/ipt/resource?r=trh_vxl http://dev.gbif.no/ipt/resource?r=trh_vxl http://gbif.no/ipt/resource?r=trh_vxl http://gbif.no/ipt/resource.do?r=trh_vxl

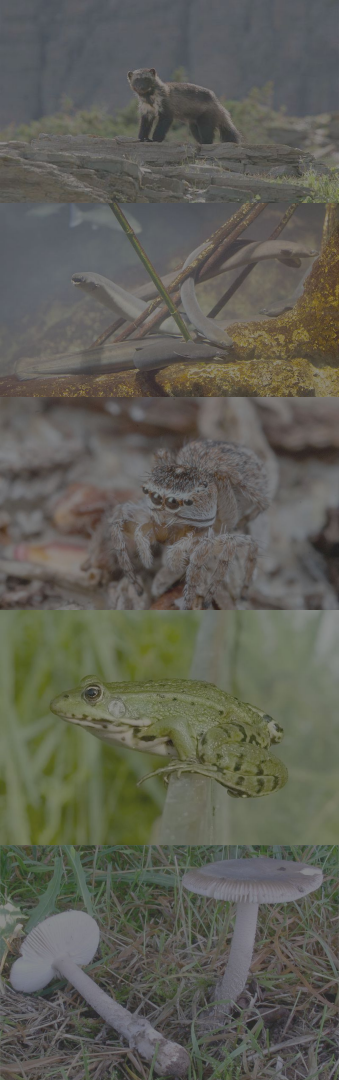
GBIF registration details

Citation

Norwegian University of Science and Technology (2025). Vascular plant field notes, NTNU University Museum. Version 1.12. Sampling event dataset https://doi.org/10.15468/kkb2x0 accessed via GBIF.org on 2026-10-19.

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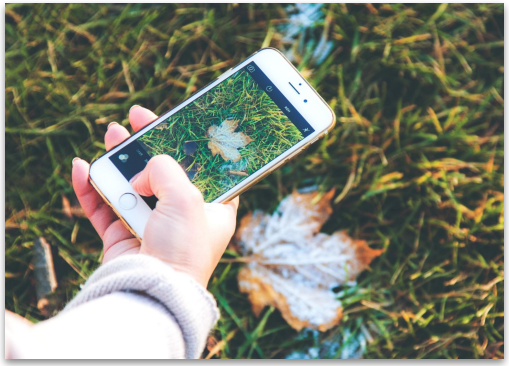
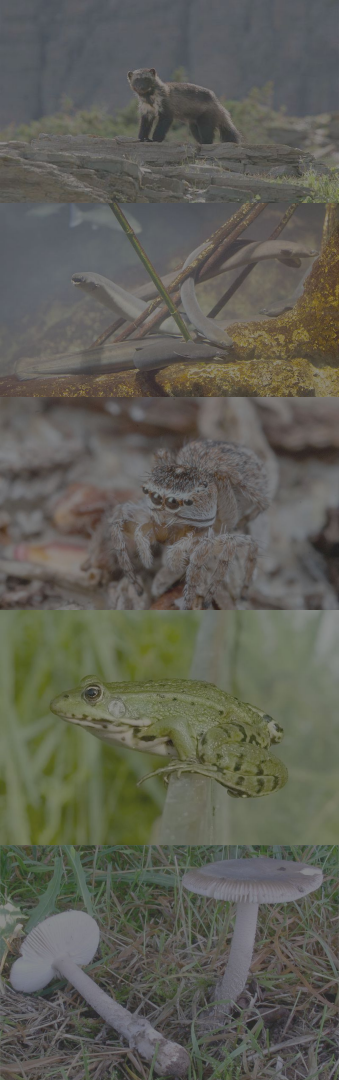
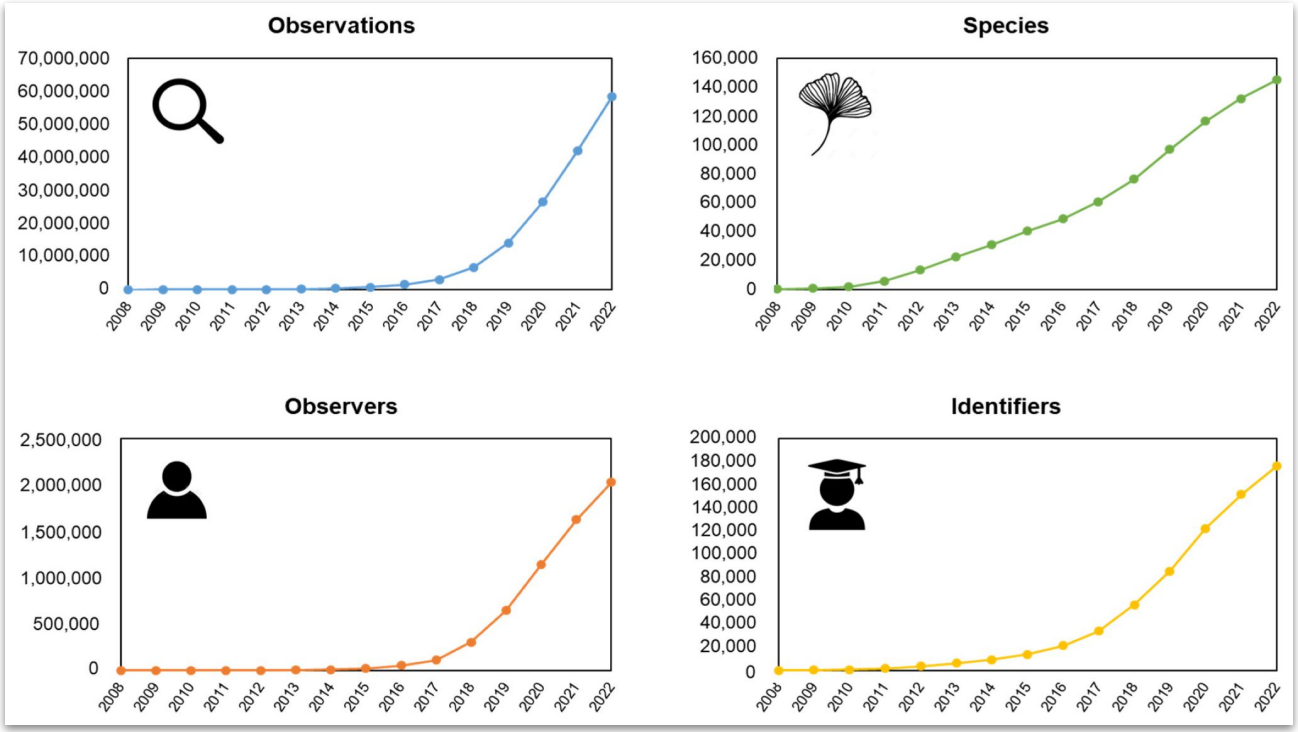


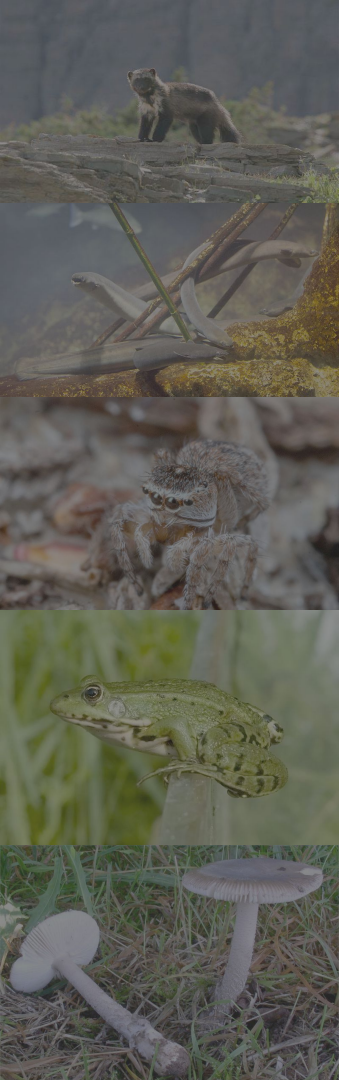


Image credit: Test tube - University of Michigan SEAS (CC BY 2.0), transect - Northwest College Agriculture (CC BY 2.0)

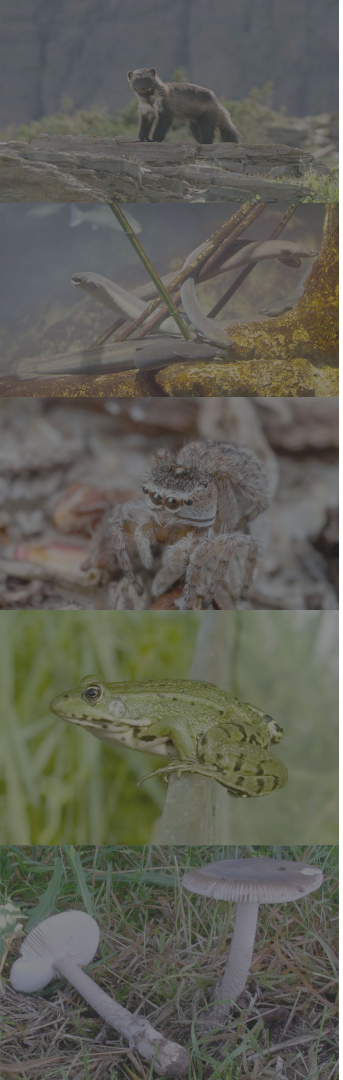




Lopez-Guillen et al, 2024



Absences get removed!



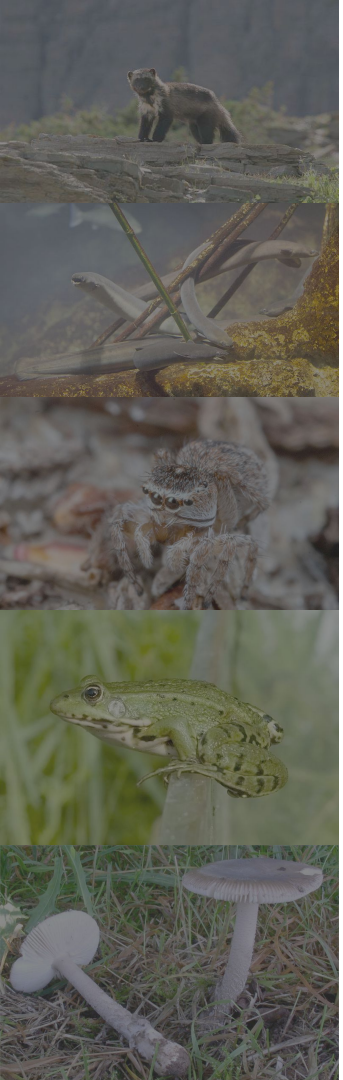
Occurrence only data is everywhere, but doesn’t have much sampling information

Presence-absence data gives you a lot more information, but it’s much harder to come by









Dag-Inge Øien Administrative point of contact

GBIF registration

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Metadata last modified	7 November 2025
Publication date	6 November 2025
Hosted by	Norwegian University of Science and Technology
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Preferred identifier	DOI 10.15468/kkb2x0
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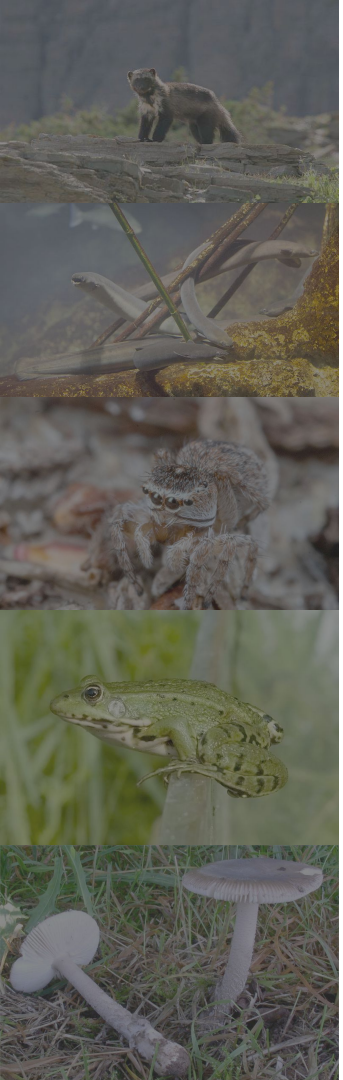
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We’ll learn how to examine GBIF datasets (and metadata) through R

We’ll download GBIF data from GBIF (and from its IPT source)

And we’ll add absence data back in, where possible